Object Oriented Programming Using C++

1. C++ was originally developed by
(a) Clocksin and Mellish (b) Donald E. Knuth
(c) Sir Richard Hadlee (d) Bjarne Stroustrup

***2.** `cfront`
(a) is the front end of a C compiler
(b) is the pre-processor of a C compiler
(c) is a tool that translates a C++ code to its equivalent C code
(d) none of the above

***3.** The following program fragment

```
int i=10;
void main( )
{
    int i=20;
    {
       int i=30.;
       cout << i << ::i;
    }
}
```

(a) prints `3010` (b) prints `3020`
(c) will result in a run time error (d) none of the above

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***4.** Which of the following are procedural languages?
(a) Pascal (b) Smalltalk (c) C++ (d) C

***5.** A function `abc` is defined as

```
void abc(int x=0, int y=0)
{ cout << x << y; }
```

Which of the following function calls is/are illegal? (Assume `h,g` are declared as integers)
(a) `abc();` (b) `abc(h);` (c) `abc(h,h);` (d) None of the above

6. The following C++ code results in

```
#include "iostream.h"
void main(void)
{
        cout << (int i=5) << (int j=6);
}
```

(a) compilation error
(b) run time error
(c) link time error
(d) none of the above

***7.** Reusability is a desirable feature of a language as it
(a) decreases the testing time (b) lowers the maintenance cost
(c) reduces the compilation time (d) reduces the execution time

***8.** Choose the correct statements regarding `inline` functions.
(a) It speeds up execution (b) It slows down execution
(c) It increases the code size (d) It decreases the code size

9. If many functions have the same name, which of the following information, if present, will be used by the compiler to invoke the correct function to be used?
(a) The operator : : (b) The return value of the function
(c) Function signature (d) None of the above

***10.** The statements

```
int a = 5;
cout << "FIRST" << (a<<2) << "SECOND";
```

outputs

(a) FIRST52SECOND (b) FIRST20SECOND
(c) SECOND25FIRST (d) an error message

11. Choose the correct remarks.

(a) C++ allows any operator to be overloaded.
(b) Some of the existing operators cannot be overloaded.
(c) Operator precedence cannot be changed.
(d) All of the above.

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12. A constructor is called whenever

(a) an object is declared (b) an object is used
(c) a class is declared (d) a class is used

***13.** Which of the following remarks about the differences between constructors and destructors are correct?

(a) Constructors can take arguments but destructors cannot.
(b) Constructors can be overloaded but destructors cannot be overloaded.
(c) Destructors can take arguments but constructors cannot.
(d) Destructors can be overloaded but constructors cannot be overloaded.

***14.** The following program fragment

```
void main( )
{
    int x=10;
    int &p=x;
    cout << &p << &x;
}
```

(a) prints 10 and the address of x (b) results in a run time error
(c) prints the address of x twice (d) prints the address of p twice

***15.** The declaration

```
int x; int &p=x;
```

is same as the declaration

```
int x,*p; p=&x;
```

This remark is

(a) true (b) false (c) sometimes true (d) none of the above

16. The following program segment

```
const int m=10;
int &n=m;
n=11;
cout << m << n;
```

(a) results in compile time error (b) results in run time error
(c) prints 1111 (d) prints 1011

***17.** The following program segment

```
int a=10;
int const &b=a;
a=11;
cout << a << b;
```

(a) results in compile time error (b) results in run time error
(c) prints 1111 (d) none of the above

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The next two questions are based on the following program segment.

```
class A
{ int i1;
   protected:
    int i2;
   public:
    int i3;
} ;
class B : public A
{ public:
  int i4;
};
class C : B
{};
```

48. The variable `i2` is accessible

(a) to a public function in `class A`
(b) to a public function in `class B`
(c) to a public function in `class C`
(d) from the main function

49. Which variable(s) is/are accessible from the main function?

(a) `i1` (b) `i2` (c) `i3` (d) None of the above

***50.** The following program

```
class abc;
class def
   {   int i1;              // statement 1
       protected: int i2;   // statement 2
       public: int i3;      // statement 3
   friend abc;
   };
   class abc
   { public:
       void main(def A)
       {cout << (A.i1=3); cout << (A.i2=4); cout << (A.i3=5)}
   );
```

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```
void main( )
{
      def x1;
      abc x2;
      x2.mn(x1);
}
```

(a) will compile successfully if statement 1 is removed
(b) will compile successfully if statement 2 is removed
(c) will compile successfully if statement 3 is removed
(d) will run successfully and print 345

53. Consider the declarations

char a;
const char aa = 'h';
char *na;

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const char *naa;

Which of the following statements

Statement I: aa = a;

Statement II: na = &a;

Statement III: na = &aa;

is/are illegal?

(a) Only I and II (b) Only II and III

(c) Only I and III (d) All the three statements are illegal

54. Forgetting to include a file (like cmath or math.h) that is necessary will result in

(a) compilation error (b) warning when the program is run

(c) error at link time (d) warning when the program is compiled

***55.** Assume that the random number generating function – rand( ), returns an integer between 0 and 10000 (both inclusive). If you want to simulate the throwing of a die using this random function, use the expression

(a) rand( ) % 6 (b) rand( ) % 6 + 1

(c) rand( ) % 5 + 1 (d) none of the above

***56.** Assume that the random number generating function – rand( ), returns an integer between 0 and 10000 (both inclusive). To randomly generate a number between a and b (both inclusive), use the expression

(a) rand( ) % (b-a) (b) (rand( ) % a) + b

(c) (rand( ) % (b-a)) + a (d) (rand( ) % (b-a+1)) + a

57. Which of the following comments about inline comments are true?

(a) A function is declared inline by typing the keyword *inline* before the return value of the function.

(b) A function is declared inline by typing the keyword *inline* after the return value of the function.

(c) A function that is declared inline may not be treated inline.

(d) Inline functions are essentially same as implementing a function as macro.

58. Which of the following decides if a function that is declared inline is indeed going to be treated inline in the executable code?

(a) Compiler (b) Linker (c) Loader (d) Preprocessor

***59.** Which of the following type of functions is an ideal candidate for being declared *inline*?

(a) A function that is small and is not called frequently.

(b) A function that is small and is called frequently.

(c) A function that is not small and is not called frequently.

(d) A function that is not small and is called frequently.

***60.** One of the disadvantages of pass-by reference is that the called function may inadvertently corrupt the caller's data. This can be avoided by

(a) passing pointers (b) declaring the formal parameters constant

(c) declaring the actual parameters constant (d) all of the above

64. A possible output of the following program fragment

```
for(i = getchar();; i = getchar())
if(i == 'x') break;
else putchar(i);
```

is

(a) mi (b) mix (c) mixx (d) none of the above

65. The following program

```
main()
{
  int i = 5;
  if (i == 5) return;
  else printf("i is not five");
  printf("over");
}
```

results in

(a) a syntax error (b) an execution error

(c) printing of over (d) execution termination, without printing anything

***66.** The following program fragment

```
int i = 5;
do {putchar(i + 100); printf("%d", i--);}
while(i);
```

results in the printing of

(a) i5h4g3f2el (b) i4h3g2fle0

(c) an error message (d) none of the above

***67.** The following program fragment

```
int i = 107, x = 5;
printf((x > 7)? "%d": "%c", i);
```

results in

(a) an execution error (b) a syntax error

(c) printing of k (d) none of the above

***68.** Replacing > by < in the previous question results in

(a) printing of 107 (b) a syntax error

(c) printing of k (d) none of the above.

***69.** The following loop

```
while(printf("%d", printf("az")))
printf("by");
```

(a) prints azbybybyby... (b) prints azbyazbyazbyazby...

(c) results in a syntax error (d) none of the above

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70. The following statements

```
for(i = 3; i < 15; i += 3)
{ printf("%d ", i);
  ++i;
}
```

will result in the printing of

(a) 3 6 9 12 (b) 3 6 9 12 15 (c) 3 7 11 (d) 3 7 11 15

71. If a = 9, b = 5 and c = 3, then the expression `(a - a/b * b%c) > a%b%c` evaluates to

(a) true (b) false (c) invalid (d) 0

72. In a `for` loop, if the condition is missing, then,

(a) it is assumed to be present and taken to be false

(b) it is assumed to be present and taken to be true

(c) it results in a syntax error

(d) execution will be terminated abruptly

73. In a `for` loop, if the condition is missing, then infinite looping can be avoided by a

(a) `continue` statement (b) `goto` statement

(c) `return` statement (d) `break` statement